

CALIBRATION POLICY

WATR — Watering Automation Twin for
Rehydration
Microclimate Coefficient Calibration · v2.0

Version	2.0	Date	June 2026
Status	Draft	Owner	Digital Twin Team
Programme	WATR	Domain	Microclimate Coefficient (μ)

1. Background

WATR predicts soil moisture levels for each pot in the balcony irrigation network. The central model parameter is the **balcony microclimate correction coefficient (μ)**, a scalar that adjusts the open-field reference evapotranspiration (ET_0) to reflect actual balcony conditions — wind shelter from walls, additional heat radiation from surfaces, and partial shading from overhangs.

The correction enters the crop evapotranspiration formula as:

$$ET_{crop} = \mu \times K_c \times ET_0$$

WATR publishes its predicted soil moisture to the **anomaly detection group**, which uses these predictions as a drift-free reference to identify sensor faults. For this reason, the integrity of μ — and therefore of the predictions — is critical. A μ that has absorbed sensor drift rather than true microclimate change would corrupt the anomaly detection group's reference signal.

2. Calibration Methods

WATR defines two calibration methods. They differ in what signal they fit μ against and in their robustness to sensor drift.

HARD
CALIBRATION

2.1 Hard Calibration

Hard calibration fits μ against **absolute soil moisture readings**. Starting from the default value μ = 1 (no correction), the model is run and μ is adjusted until predictions align with the sensor. Because it relies on absolute values, hard calibration assumes the sensor is reading correctly at the time of calibration.

Two implementation approaches:

Stepwise (online)	μ is updated incrementally after each new (input, sensor reading) sample, similar to stochastic gradient descent. The coefficient moves toward the optimum continuously as data arrives. <i>Suitable when calibration data is sparse and the system must start producing estimates quickly.</i>
Batch Regression (offline)	A window of N (input, sensor reading) sample pairs is collected first. μ is then fitted in a single least-squares regression step. Because μ enters ET_{crop} linearly, this reduces to a one-dimensional linear regression. <i>Preferred when enough data is available; produces a more stable, noise-robust estimate than stepwise updating.</i>
NOTE	Hard calibration is susceptible to sensor drift: if the sensor has an offset at calibration time, that offset will be absorbed into μ . Hard calibration should therefore only be run when sensor health is confirmed — either at system initialisation or after the anomaly detection group has resolved a drift event.

SOFT CALIBRATION

2.2 Soft Calibration

Soft calibration fits μ against the **rate of change of soil moisture** ($d\theta/dt$) rather than absolute moisture levels. This is a **first-difference estimator**: instead of comparing θ_{sensor} to $\theta_{\text{predicted}}$, it compares $\Delta\theta_{\text{sensor}}$ to $\Delta\theta_{\text{predicted}}$ over consecutive time steps.

The motivation is **drift immunity**. Sensor drift is approximately an additive, slowly varying offset on the absolute moisture reading. When the first difference is taken, this offset cancels — a sensor that reads consistently 5% high will show the same $\Delta\theta$ as a healthy sensor, because the offset appears in both the current and previous reading and subtracts out. Seasonal change in μ , by contrast, alters the *rate* at which moisture declines between irrigations and is therefore visible in $\Delta\theta$. Soft calibration can thus track seasonal microclimate drift without learning from sensor offset drift, preserving the integrity of WATR's predictions as a reference for the anomaly detection group.

Batch update mechanism:

To suppress noise in individual $\Delta\theta$ samples, soft calibration updates μ in batches rather than on every reading. A batch of a consecutive ($\Delta\theta_{\text{sensor}}$, $\Delta\theta_{\text{predicted}}$) pairs is accumulated; μ is then updated once per batch using the aggregated signal. This is repeated up to a maximum of X updates, or until the convergence criterion below is met.

Convergence criterion:

Soft calibration is considered converged — and updates stop — when the change in μ between consecutive batch updates remains within a tolerance σ for b consecutive updates. Formally:

$$|\mu_{-}(k) - \mu_{-}(k-1)| < \sigma \text{ for } b \text{ consecutive } k$$

The parameters a , X , σ , and b are configurable and should be tuned to the expected rate of seasonal μ drift and the noise level of the sensor.

NOTE	Soft calibration's drift immunity applies to <i>additive</i> sensor offset drift — the dominant failure mode for capacitive soil moisture sensors. Multiplicative (scaling) drift does affect $\Delta\theta$ and would still corrupt soft calibration. This edge case remains the responsibility of the anomaly detection group.
-------------	--

3. Calibration and Recalibration Policy

The following table defines which calibration method applies to each trigger. All implementations must conform to this mapping.

Trigger	Method	Rationale
1. Initial fitting (system first boot)	Hard Calibration	No prior μ exists. The sensor is assumed healthy at deployment. Absolute fitting from the default $\mu = 1$ establishes the baseline coefficient.
2. Seasonal / periodic recalibration	Soft Calibration	Microclimate changes slowly over weeks and seasons. Soft calibration tracks this drift via $\Delta\theta$ fitting without risk of absorbing sensor offset drift into μ , keeping WATR predictions drift-free for the anomaly detection group.
3. Physical position change	Hard Calibration	Relocating pots or system components invalidates the current μ entirely. A full hard recalibration is needed to re-establish the coefficient for the new microclimate exposure. Sensor health must be confirmed before triggering.
4. Anomaly detection group request (post drift resolution)	Hard Calibration	If a sensor drift event corrupted recent data, the current μ may have shifted. After the anomaly detection group resolves the drift and confirms sensor health, a hard recalibration re-anchors μ to clean absolute readings.
5. User-initiated recalibration	User's Choice (Soft recommended)	The user may trigger recalibration at any time. Soft calibration is recommended as it is safer in the presence of possible sensor offset. Hard calibration is available if the user has confirmed sensor health and wants a full re-anchor.

4. Apparent Calibration Need — False Positives

The following situations may appear to indicate that μ is incorrect. They should **not** trigger recalibration, as they are transient or out-of-scope phenomena.

Sudden Moisture Spike or Drop	A rapid change caused by rain, water splash, or accidental manual watering is a one-off disturbance. The prediction will converge back within a few cycles as the soil equilibrates. No μ adjustment is warranted.
Prediction Lag After Irrigation	Immediately after a pump activation, the prediction may lag behind the sensor while water distributes through the soil. This is an expected transient of the simulation step, not a coefficient error.

Sensor Drift (Absolute Offset)	A drifting sensor reads systematically too high or too low. This is the anomaly detection group's domain. Soft calibration is inherently immune to additive offset drift by design. Hard calibration must not be triggered while drift is active — wait for the anomaly detection group's resolution signal (Trigger 4).
Irrigation Transients	In the time steps immediately following pump activation, moisture readings may be unstable as water moves through the pipe network. Calibration assessment should exclude these windows.

This is an internal document of the Digital Twin Team, WATR programme. It is subject to revision as the system evolves.